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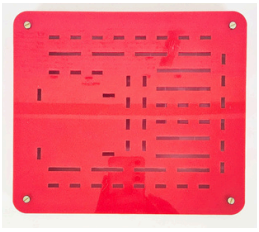
SESSION NAME

Session

10

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1 

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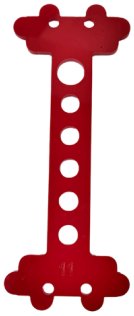
11 

12

straw

1

Spacers

2

L Clamp

3

Spacer Screws

4

Nuts

5

Screw

6

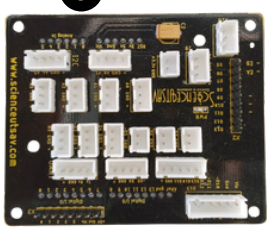
Washer

7

zinc wire 1M

8

screw hook

9 

Arduino With Shield

10

Arduino cable

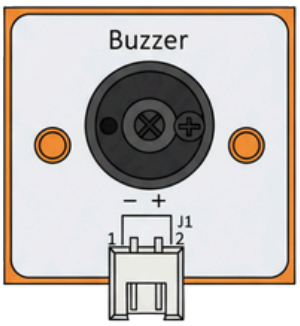
ELECTRONIC COMPONENTS

Touch sensor

A touch sensor is an electronic device that detects human touch and works like a switch. It senses the change in capacitance when a finger touches it and sends a signal to the microcontroller. This signal can be used to perform actions like turning on an LED or activating a device. Touch sensors are commonly used in mobile screens, touch lamps, and smart systems.

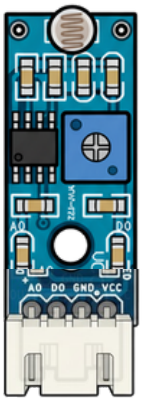
7-segment display

7 seg display electronic display device used to show numbers. It has seven LED segments arranged in the shape of the digit 8. By turning ON and OFF different segments, it can display numbers from 0 to 9. It is commonly used in digital clocks, calculators, counters, meters, and electronic device



BUZZER PCB

A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.



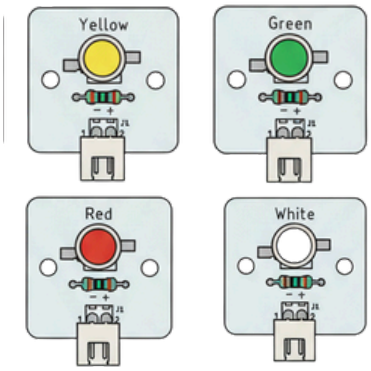
LDR sensor- A 4-pin LDR sensor is a light-sensing module that uses a Light Dependent Resistor (LDR) to detect brightness levels in the environment. It changes its resistance based on the amount of light falling on it—low resistance in bright light and high resistance in darkness. The 4-pin module typically includes extra circuitry (like a comparator) and has pins for VCC, GND, digital output (DO), and analog output (AO). It can be used to detect light intensity and trigger actions, such as turning lights on or off automatically.

ELECTRONIC COMPONENTS

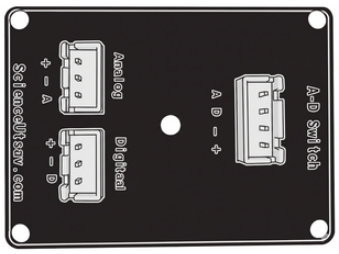
DIP LED

A DIP LED means Dual In-line Package LED — it's the common through-hole type LED you see in electronics projects.

DIP LED is a small light component that glows when electricity flows in one direction.



1W LED- A 1W LED is a high-power light-emitting diode designed to produce brighter light compared to standard LEDs. It requires more current and proper heat management, usually mounted on a small metal base for heat dissipation. It emits intense light when current flows in the correct direction and is commonly used in lighting applications like flashlights, lamps, and spotlights.

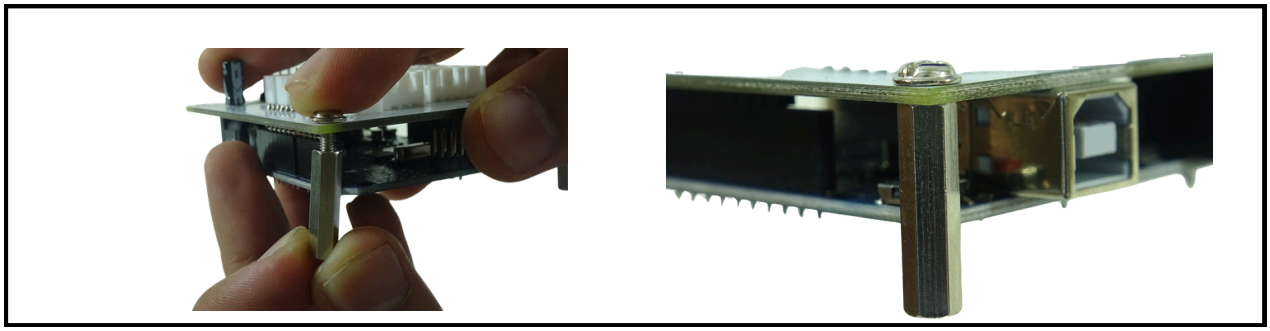


AD switch PCB- An AD switch PCB is a small circuit board that contains a switch along with basic electronic components to provide both analog and digital signal outputs. It is typically used to detect button presses or toggle actions and send signals to a microcontroller. The PCB may include pins for VCC, GND, analog output (AO), and digital output (DO), making it useful in projects where both precise (analog) and simple on/off (digital) input detection is needed.

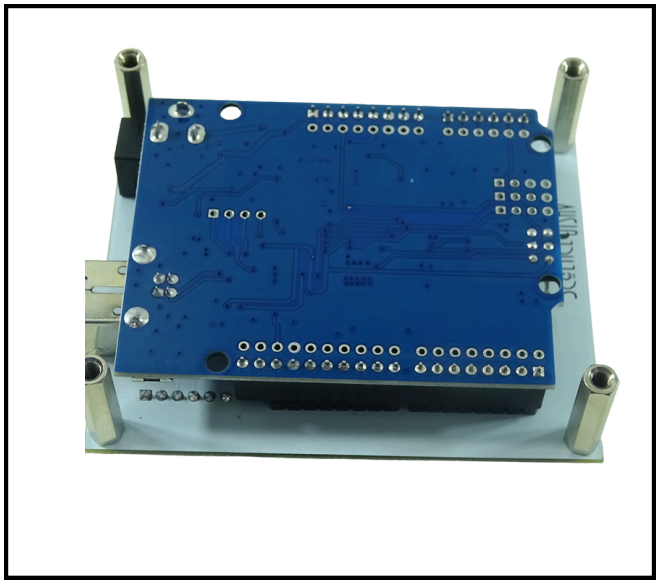
String LED- A string LED is a series of multiple LEDs connected together in a chain to produce decorative or continuous lighting. These LEDs are arranged in a sequence and powered by a single source, allowing them to glow together or in patterns. String LEDs are commonly used in applications like decorative lighting, festive displays, and ambient illumination, providing uniform light over a larger area.

Procedure :

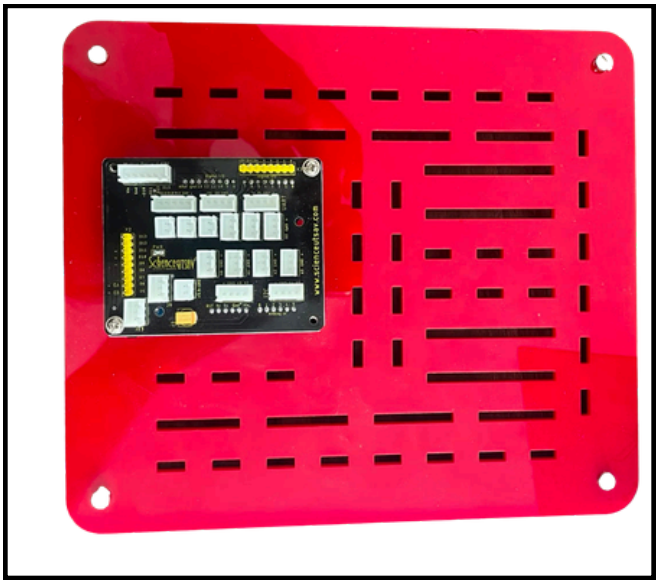
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



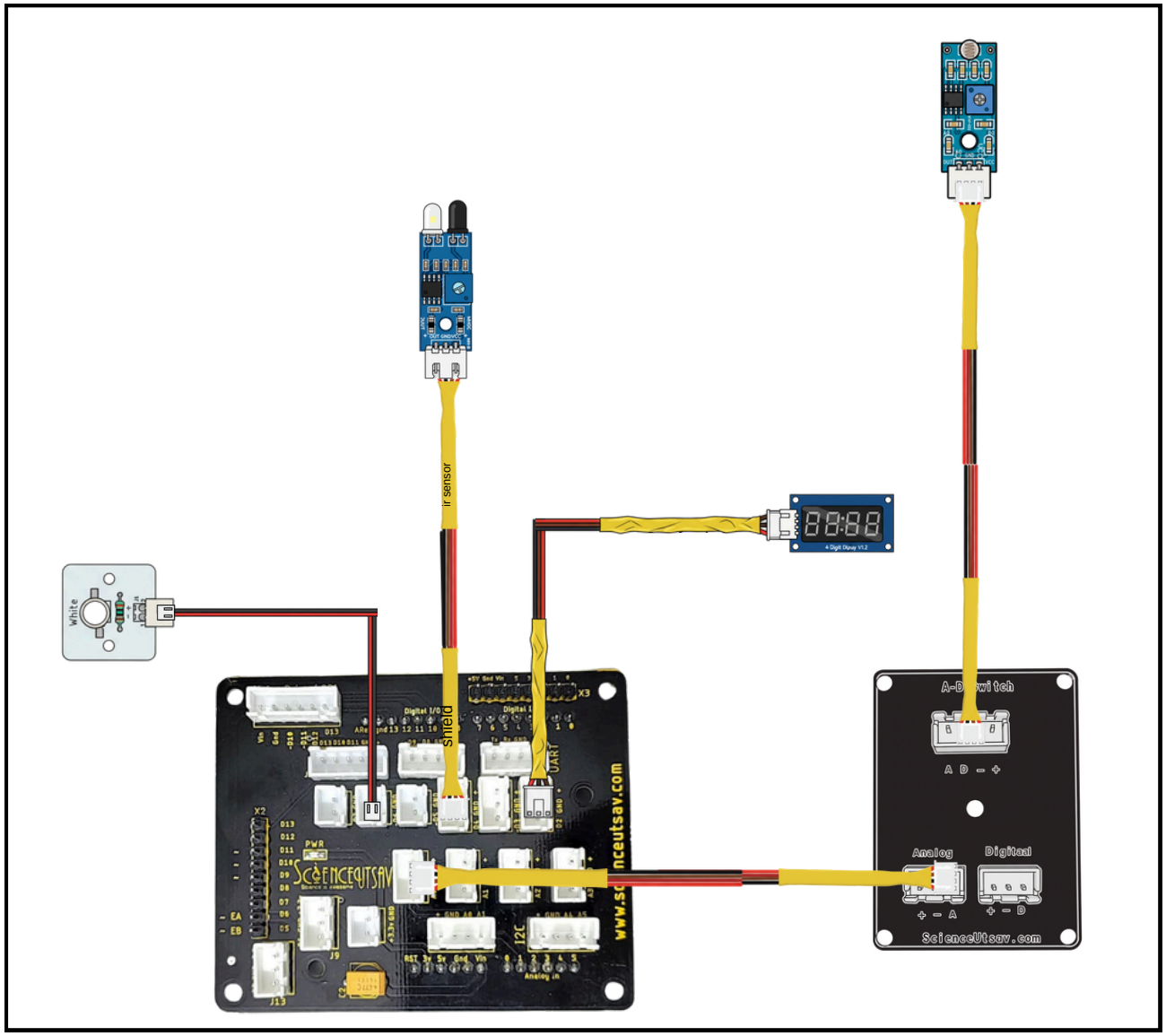
Step 3 : Place the Arduino Shield to the base board as shown below



Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

The circuit diagram of the project is given below



CONNECTIONS:

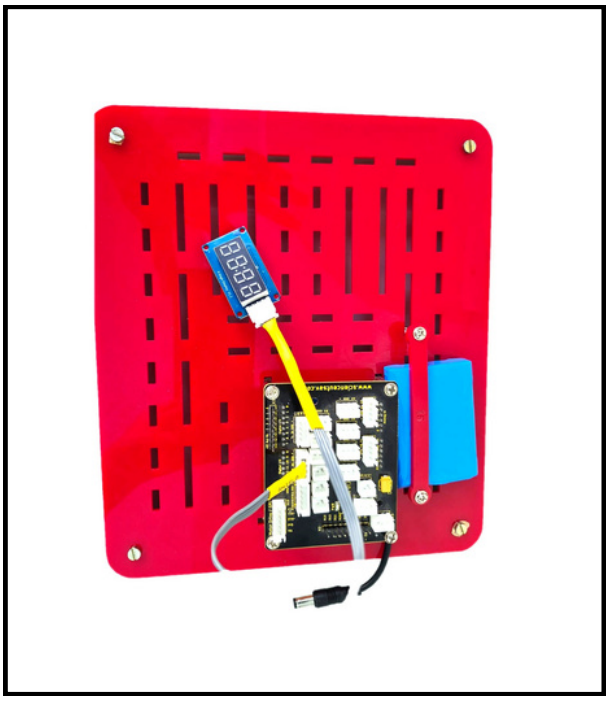
1W LED - D6

IR Sensor - D5

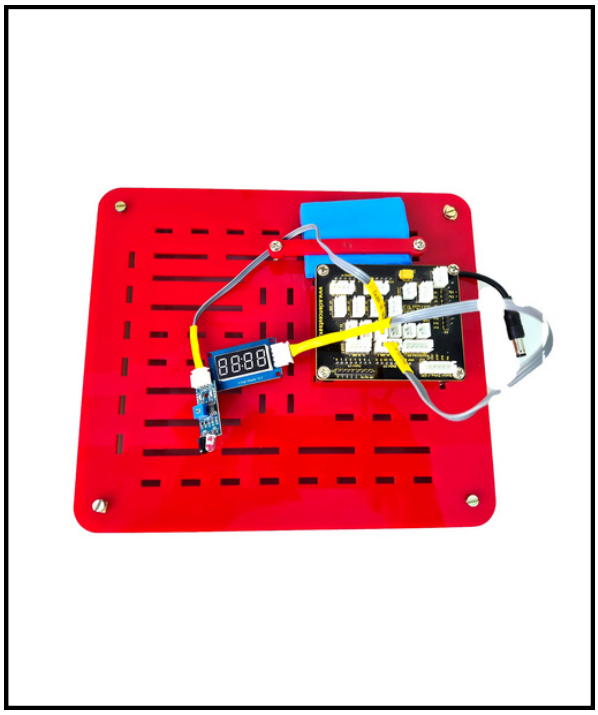
7 Segment Display - D8 D9

LDR Sensor - AD Switch A0

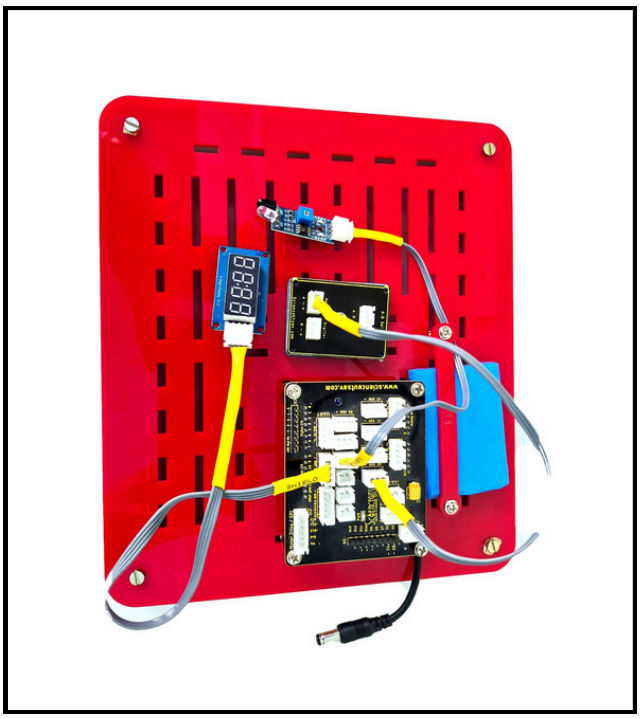
Step1- Connect the 7 segment display to the board



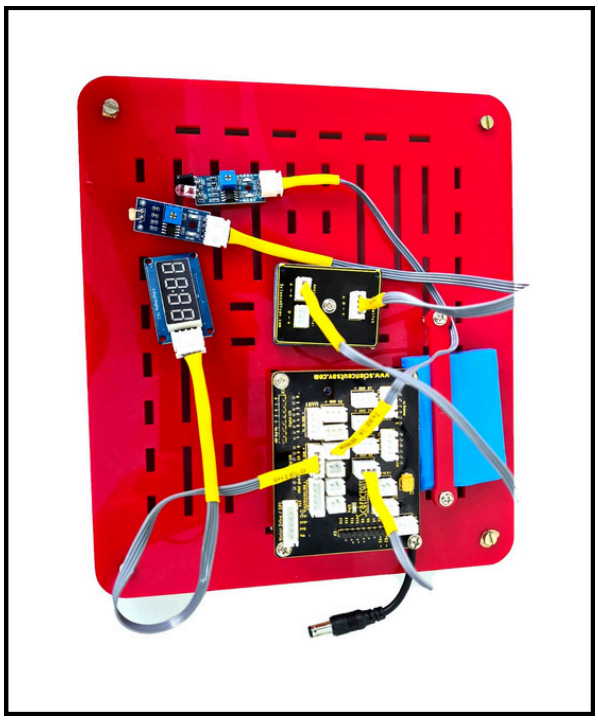
Step 2- Attach the IR sensor to the board



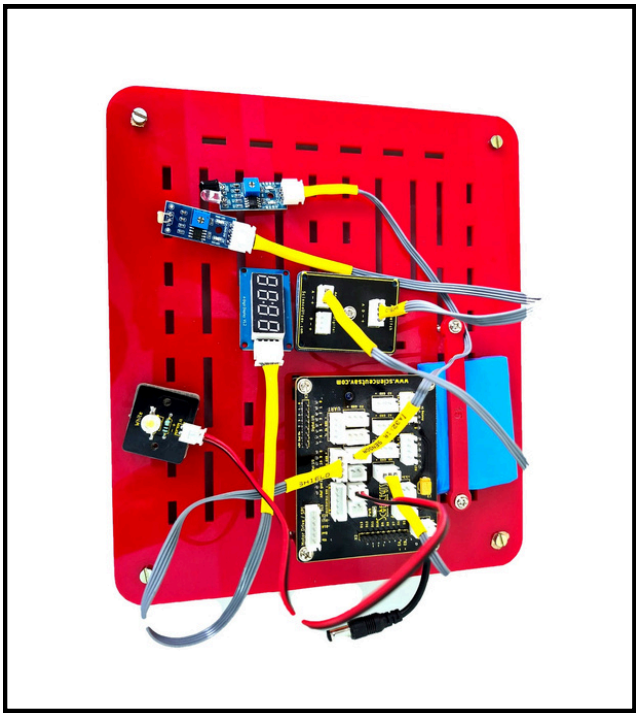
Step 3- Attach the AD switch to the base



Step 4- Connect the LDR sensor to the AD switch

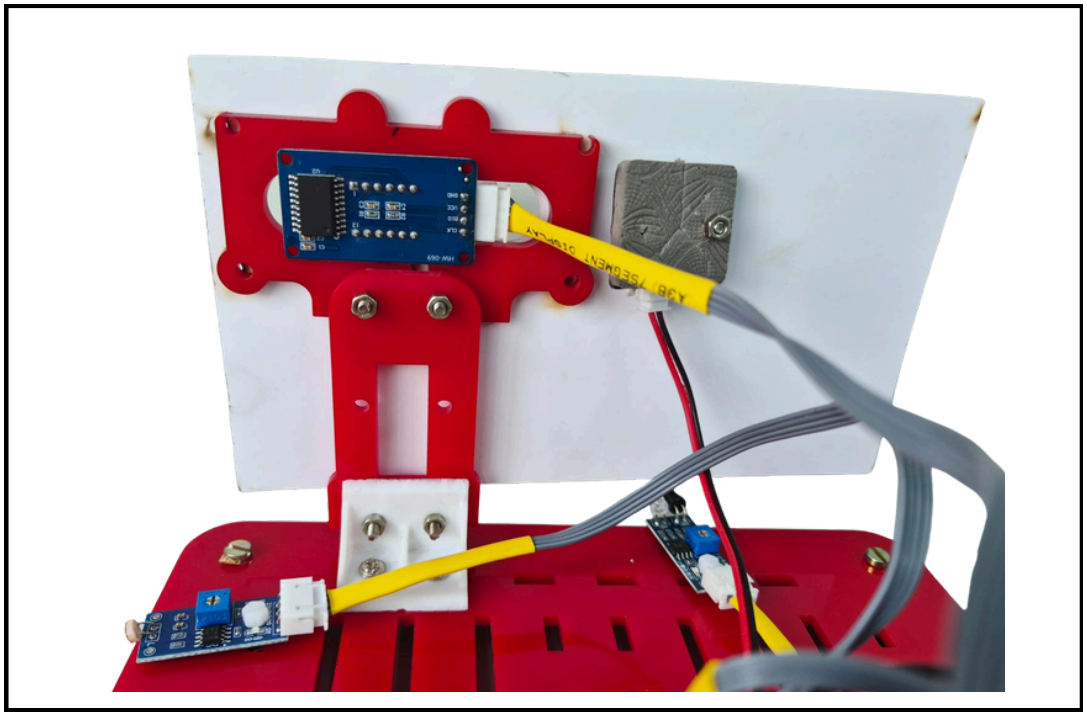


Step 5- Connect the 1W LED white light to the board as well



Step 6- Attach the template using a stand

Step 7- Place the sensors and display as shown



Step 8- Upload the code
and observe

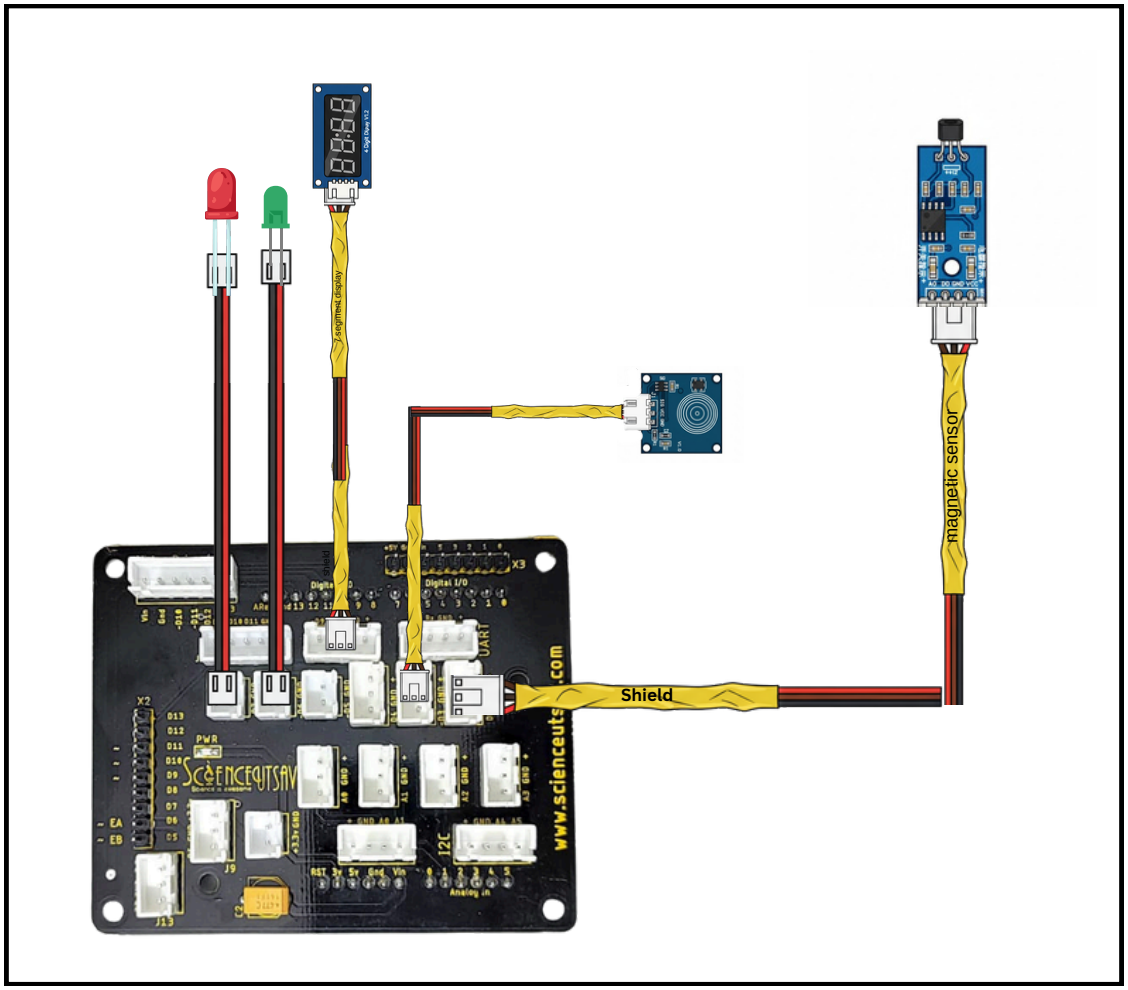
Code

```
// Session 10 Grade 6
 // Smart Home With Counter and Automatic Night Light
 // LDR controls 1W LED
 // IR counts people entering and shows count on 7 segment

 #include <TM1637Display.h>

 const int ldrSensor = A0;
 const int irSensor = 2;
 const int led = 6;
 const int CLK = 9;
 const int DIO = 8;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite
 int darkValue = 500;         // adjust based on room light

 TM1637Display display(CLK, DIO);

 int personCount = 0;
 bool lastIR = HIGH;
```

```
void setup() {
pinMode(irSensor, INPUT);
pinMode(led, OUTPUT);
display.setBrightness(7);
display.showNumberDec(0, true);
}
void loop() {
int ldrValue = analogRead(ldrSensor);
if (ldrValue > darkValue) {
digitalWrite(led, HIGH); // night light ON
} else {
digitalWrite(led, LOW); // enough light, LED OFF
}
bool currentIR = digitalRead(irSensor);
if (currentIR == IR_DETECTED && lastIR != IR_DETECTED) {
personCount++;
if (personCount > 9999) personCount = 0;
display.showNumberDec(personCount, true);
delay(300);
}
lastIR = currentIR;
}
```

BEGINNER PROJECT

The circuit diagram of the project is given below



CONNECTIONS:

7 Segment Display - D8 D9

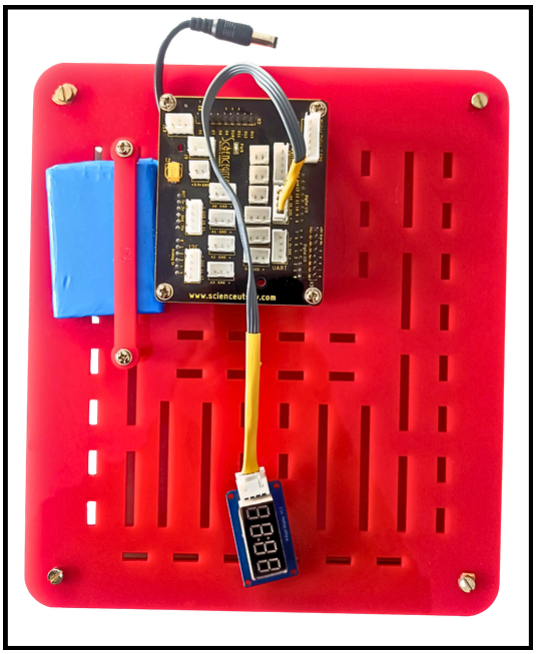
Touch Sensor - D3

Hall Effect Sensor - AD Switch A0

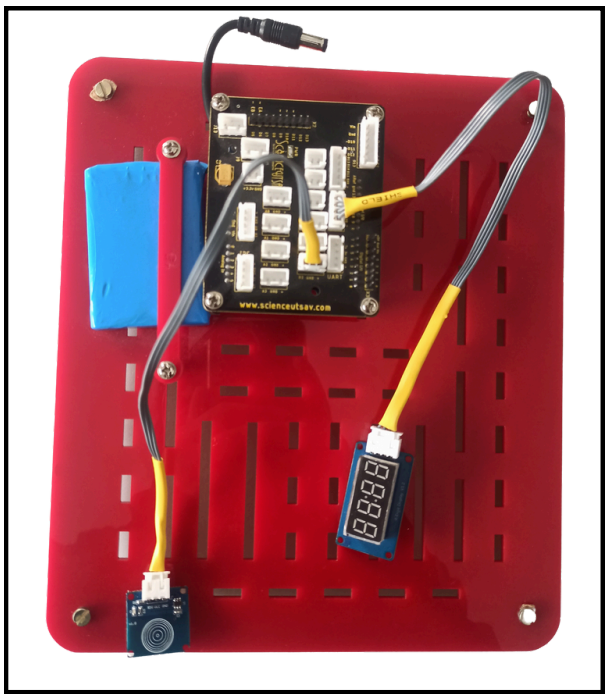
Red LED - D7

Green LED - D6

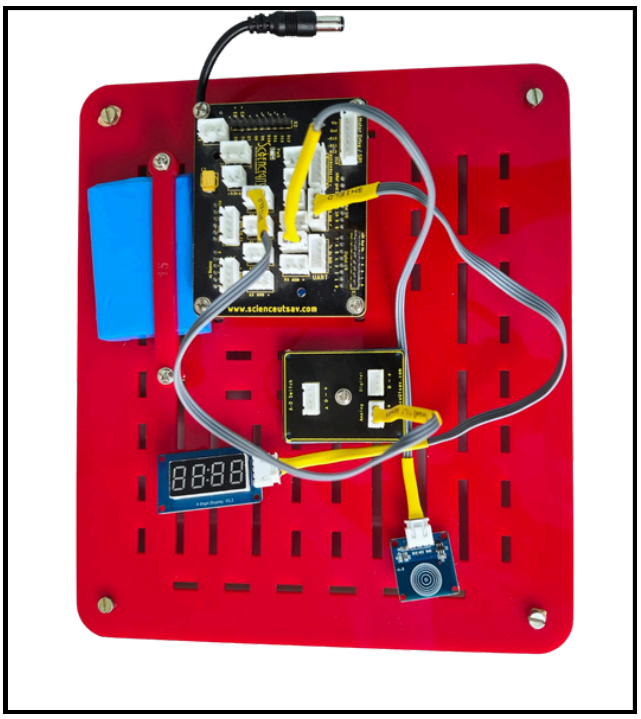
Step 1- Attach the 7 segment display to the aurdino as shown



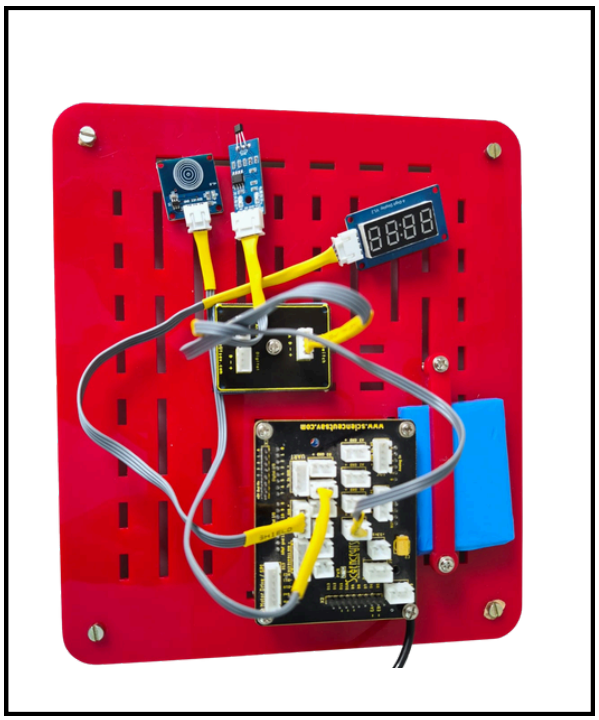
Step 2- Connect the touch sensor to the aurdino



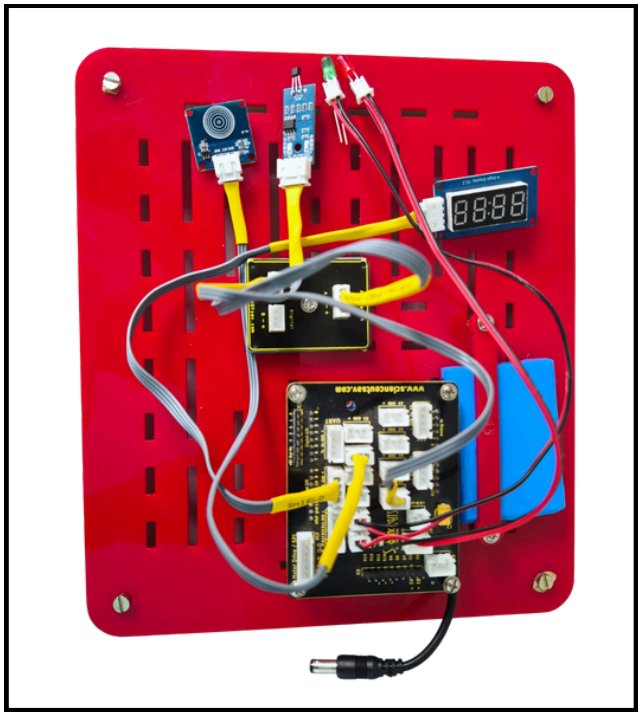
Step 3- Take 3 pin cable and connect AD switch to shield A0



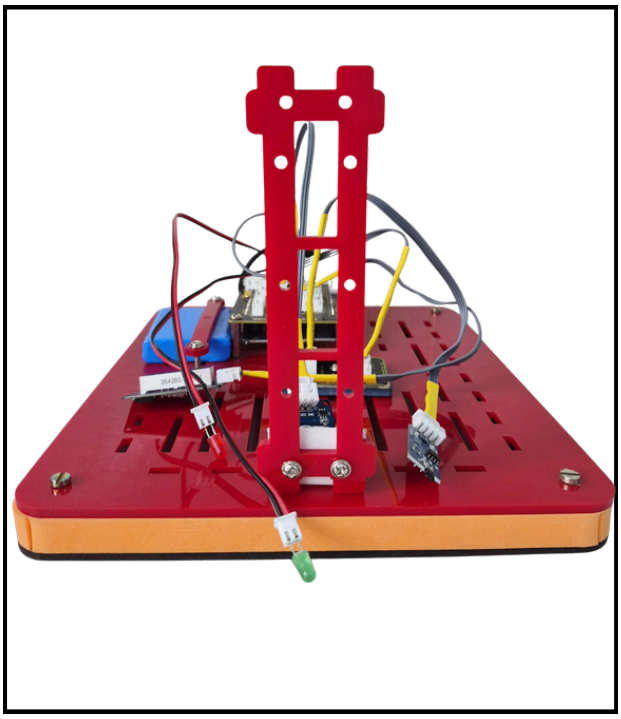
Step 4- Connect the Hall effect sensor to the AD switch



Step 5- Connect the red LED to pin D7 and the green LED to pin D6 on the Arduino shield.

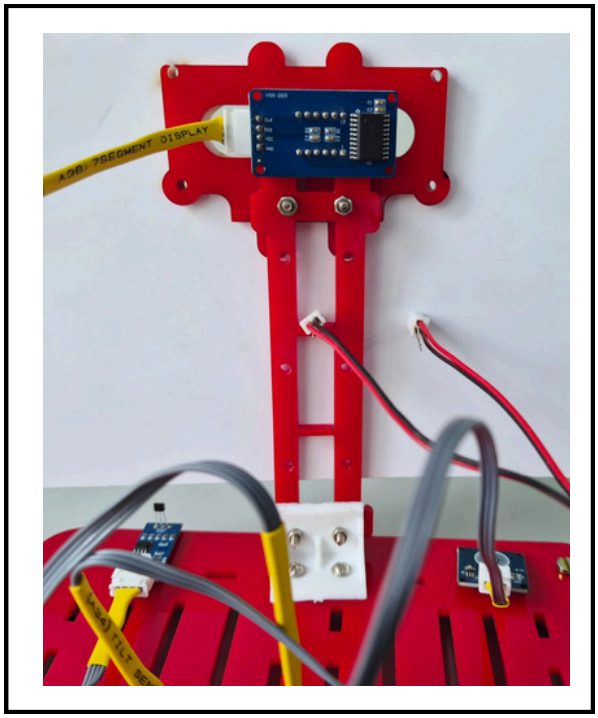


Step 6- Screw in the stand as shown



Step 7- Attach the template as shown

Step 8- Place and attach the sensors as shown with screws



Step 9- Upload the code and observe

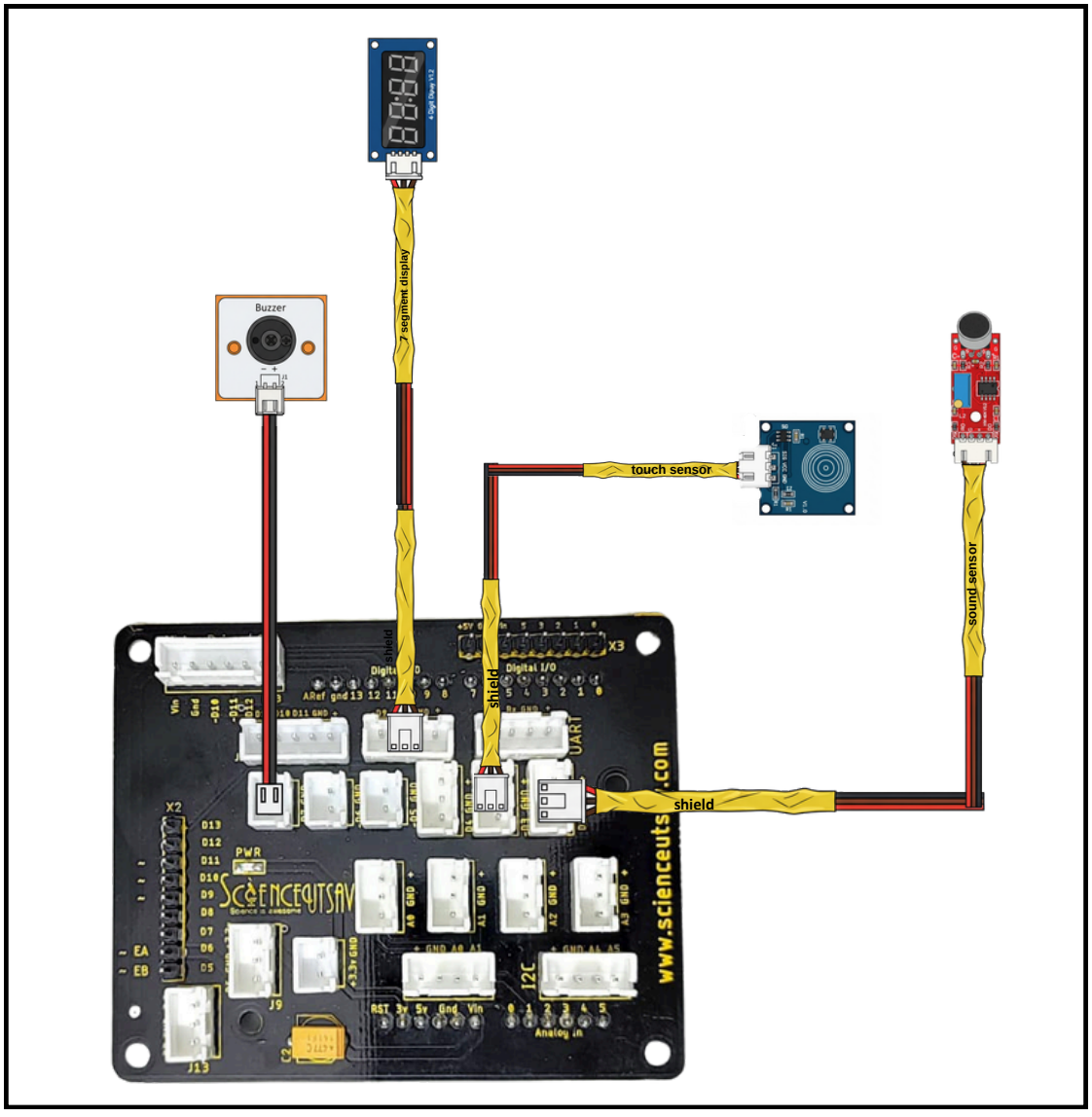
Code

```
// Session 10 Grade 7
 // Hospital Patient Call System
 // Touch sensor = doctor calls next patient, red status ON
 // Patient taps magnet card to Hall sensor = status changes to green
 // D12 is kept always HIGH for RGB project
 // Hall sensor is used as DIGITAL sensor with digitalRead(A0)

 #include <TM1637Display.h>

 const int touchSensor = 2;
 const int hallSensor = A0;
 const int redPin = 10;
 const int greenPin = 11;
 const int bluePin = 12;
 const int CLK = 9;
 const int DIO = 8;

 const int TOUCH_ACTIVE = HIGH;     // change to LOW if opposite
 const int MAGNET_DETECTED = LOW;  // change to HIGH if opposite

 TM1637Display display(CLK, DIO);

 int patientNumber = 0;
 bool patientInside = false;
 bool lastTouch = LOW;
 bool lastMagnet = HIGH;

 void setup() {
 pinMode(touchSensor, INPUT);
 pinMode(hallSensor, INPUT);
 pinMode(redPin, OUTPUT);
 pinMode(greenPin, OUTPUT);
 pinMode(bluePin, OUTPUT);

digitalWrite(bluePin, HIGH); // D12 always HIGH
 display.setBrightness(7);
 display.showNumberDec(0, true);
```

```
showGreenStatus();
}

void loop() {
digitalWrite(bluePin, HIGH); // keep D12 always HIGH

bool touchNow = digitalRead(touchSensor);
bool magnetNow = digitalRead(hallSensor);

if (touchNow == TOUCH_ACTIVE && lastTouch != TOUCH_ACTIVE) {
patientNumber++;
if (patientNumber > 9999) patientNumber = 1;
patientInside = true;
display.showNumberDec(patientNumber, true);
showRedStatus();
delay(300);
}

if (magnetNow == MAGNET_DETECTED && lastMagnet != MAGNET_DETECTED) {
patientInside = false;
showGreenStatus();
delay(300);
}

lastTouch = touchNow;
lastMagnet = magnetNow;
}

void showRedStatus() {
digitalWrite(redPin, HIGH);
digitalWrite(greenPin, LOW);
digitalWrite(bluePin, HIGH);
}

void showGreenStatus() {
digitalWrite(redPin, LOW);
digitalWrite(greenPin, HIGH);
digitalWrite(bluePin, HIGH);
}
```

INTERMEDIATRE PROJECT

The circuit diagram of the project is given below



CONNECTIONS:

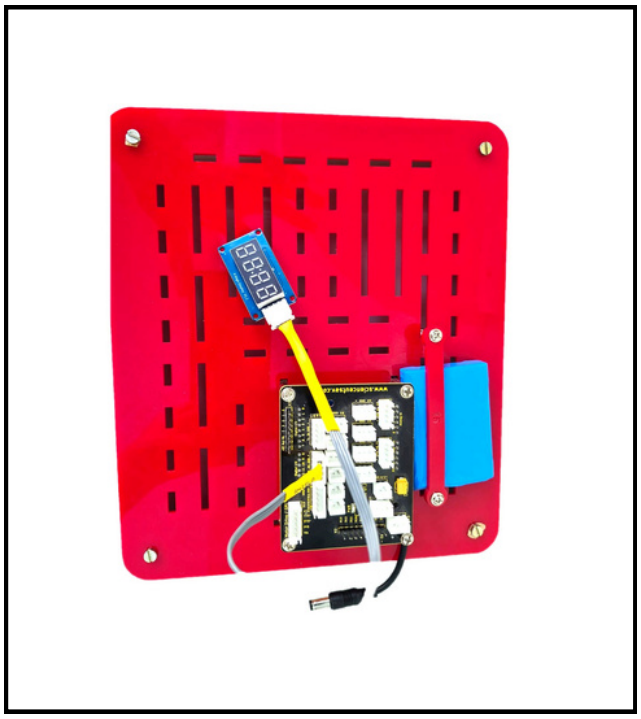
Touch Sensor - D3

Buzzer - D7

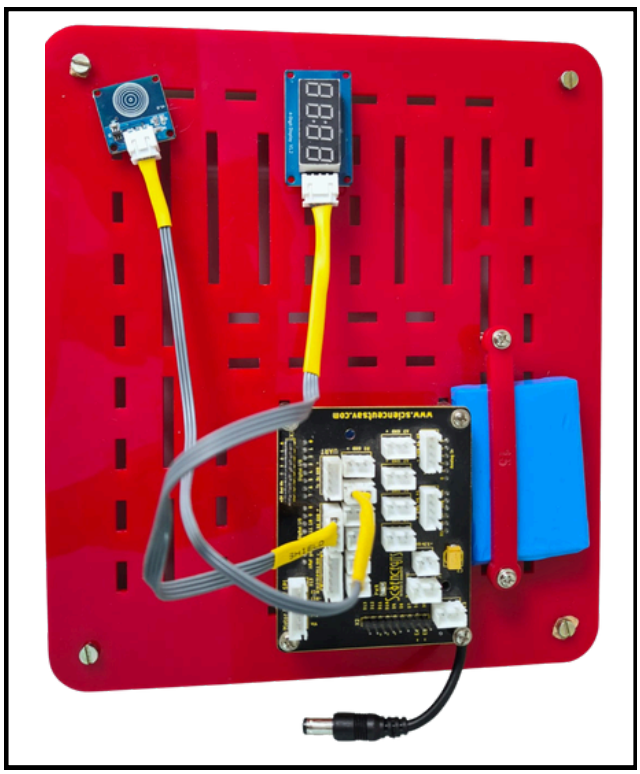
7 Segment Display - D8 D9

Sound Sensor - D2

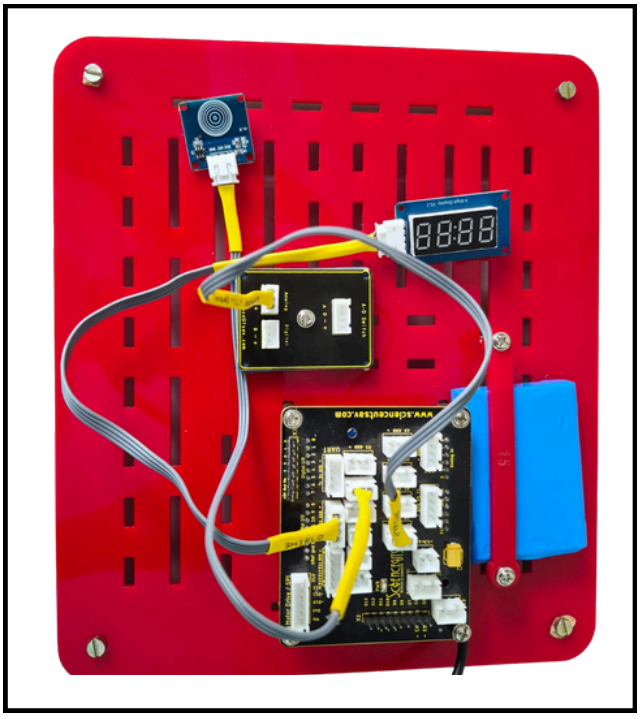
Step 1- Connect the 7 segment display to the board



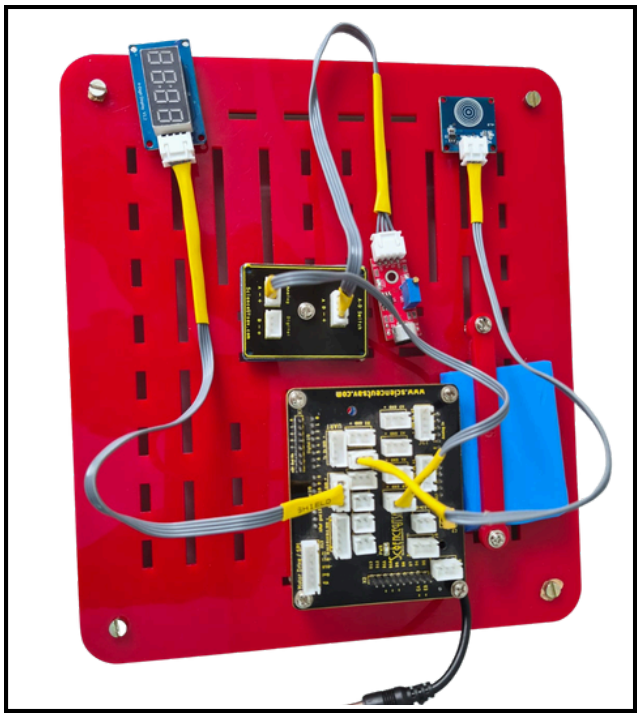
Step 2- Attach the touch sensor to the board



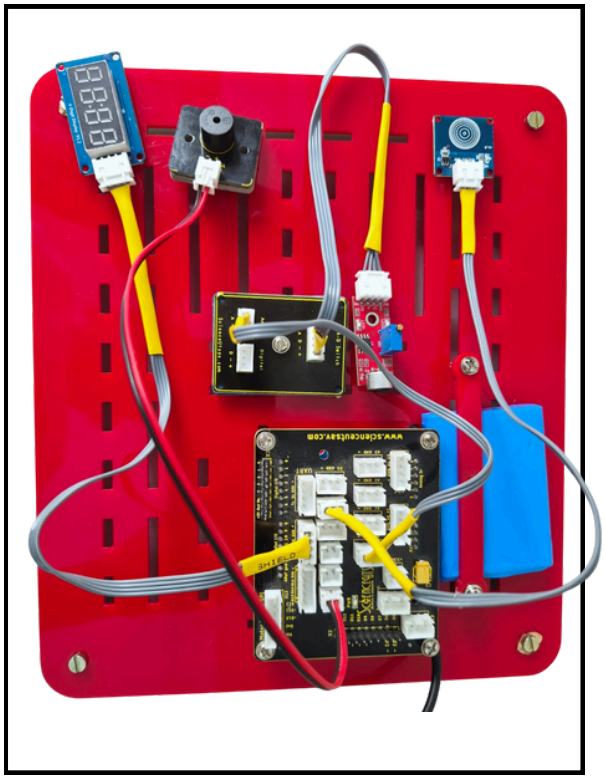
Step 3- Take 3 pin cable and connect AD switch to shield A0



Step 4- Connect the Sound sensor to the AD switch



Step 5- Connect the buzzer to the board



Step 6- Screw in the stand as shown

Step 7- Attach the template as shown

Step 8- Place and attach the sensors as shown with screws

Step 9- Upload the code and observe

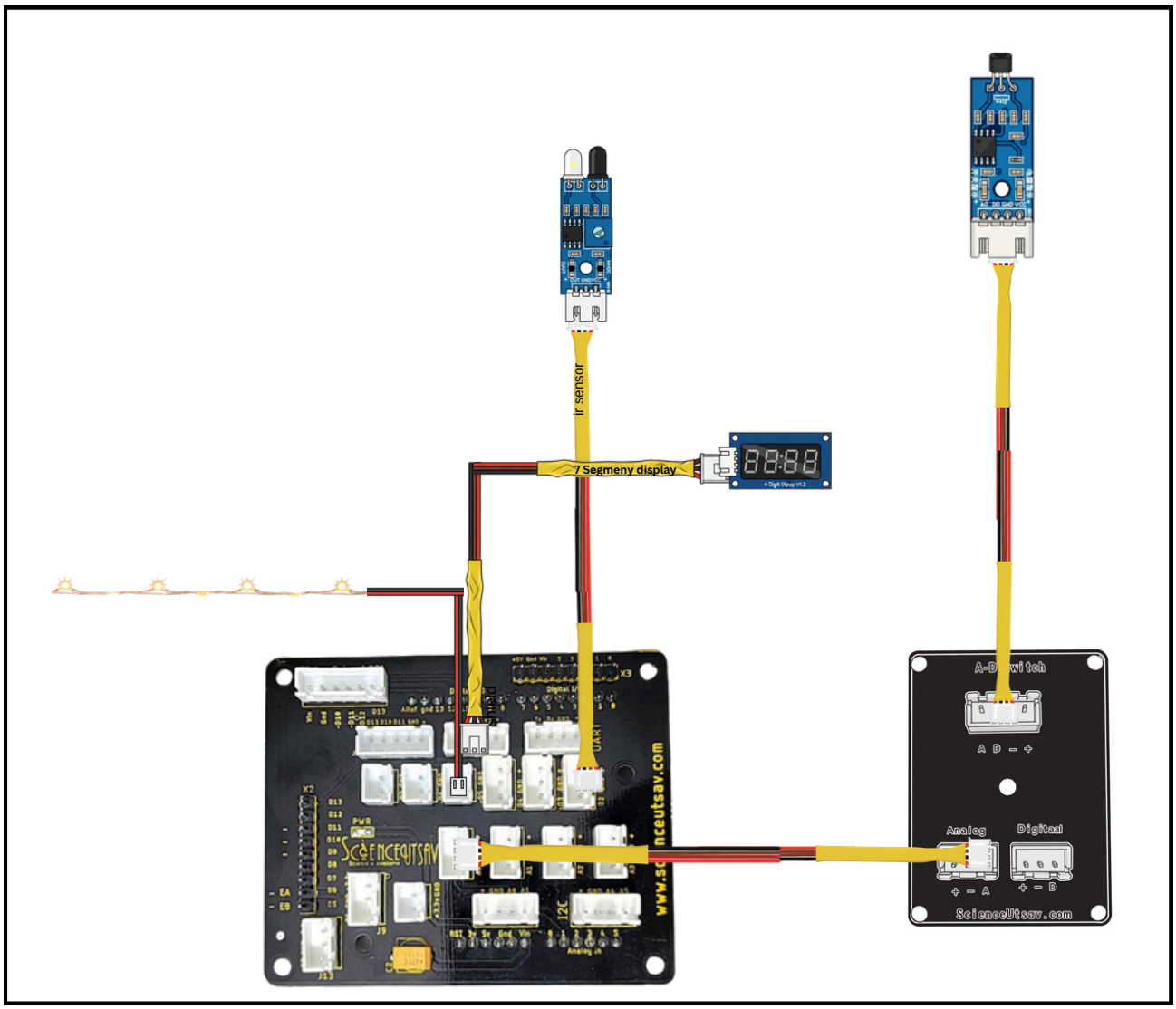
Code

```
// Session 10 Grade 8
 // Classroom Noise Meter + Emergency System
 // Sound sensor value shown on 7 segment
 // High noise = buzzer warning
 // Touch sensor = emergency buzzer pattern

 #include <TM1637Display.h>

 const int soundSensor = A0;
 const int touchSensor = 2;
 const int buzzer = 7;
 const int CLK = 9;
 const int DIO = 8;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite
 int noiseLimit = 700;

 TM1637Display display(CLK, DIO);

 void setup() {
 pinMode(touchSensor, INPUT);
 pinMode(buzzer, OUTPUT);

 display.setBrightness(7);
 display.showNumberDec(0, true);
 }

 void loop() {
 int soundValue = analogRead(soundSensor);
 int noiseLevel = map(soundValue, 0, 1023, 0, 9999);
 display.showNumberDec(noiseLevel, true);
```

```
if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
emergencyAlarm();
} else if (soundValue > noiseLimit) {
digitalWrite(buzzer, HIGH);
delay(100);
digitalWrite(buzzer, LOW);
delay(100);
} else {
digitalWrite(buzzer, LOW);
}
}
void emergencyAlarm() {
for (int i = 0; i < 5; i++) {
digitalWrite(buzzer, HIGH);
delay(150);
digitalWrite(buzzer, LOW);
delay(150);
}
}
```

EXPERT PROJECT

The circuit diagram of the project is given below



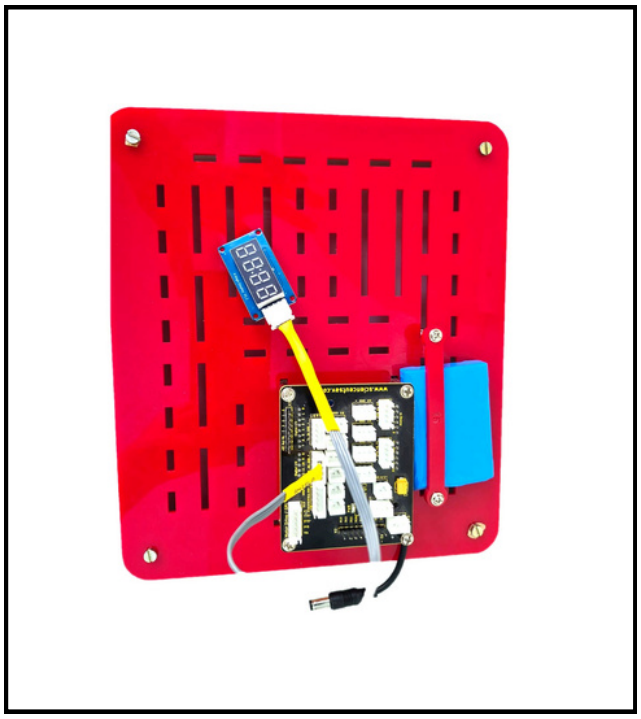
CONNECTIONS:

7 Segment Display - D8 D9 IR Sensor - AD Switch A0

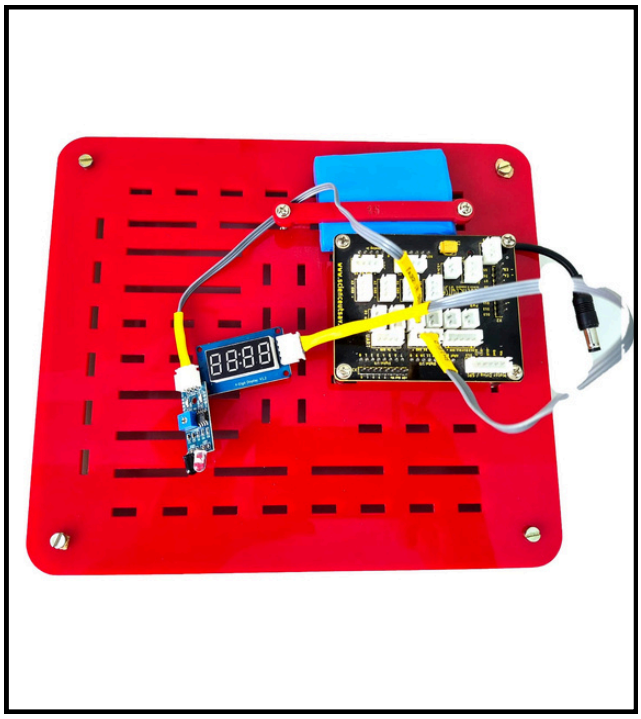
String LED - D5

Hall Effect Sensor - D4

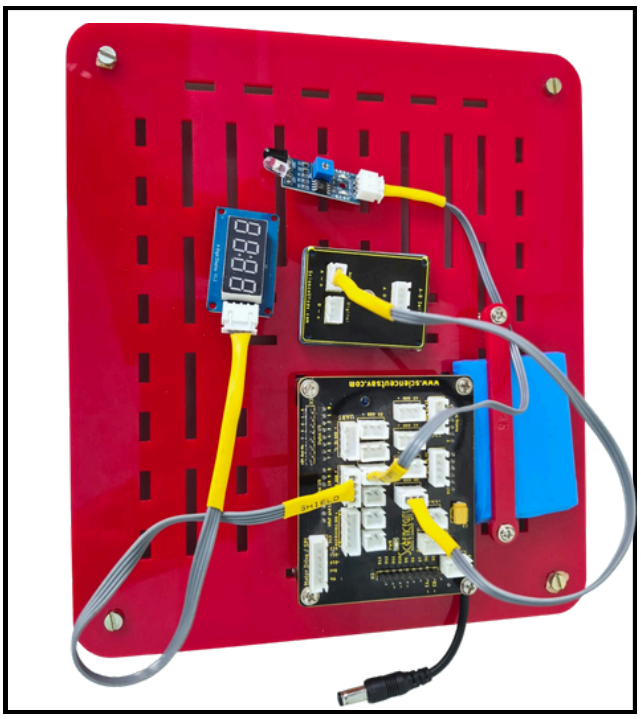
Step 1- Connect the 7 segment display to the board



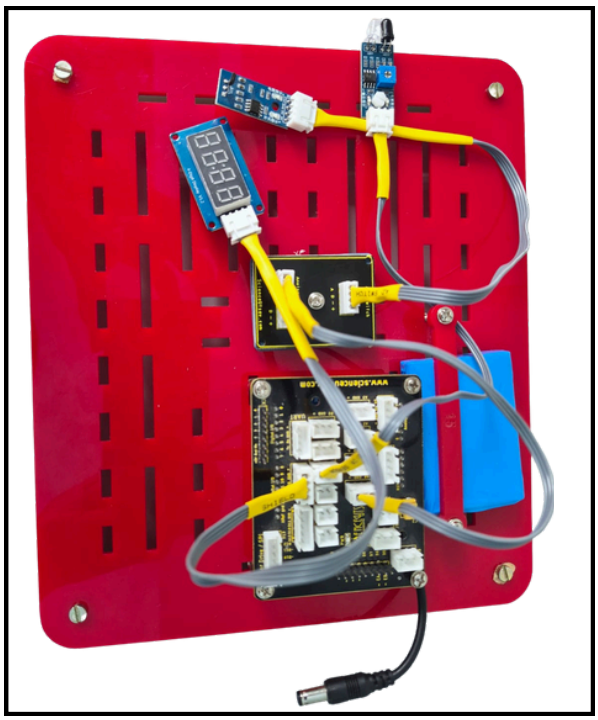
Step 2- Attach the IR sensor to the board



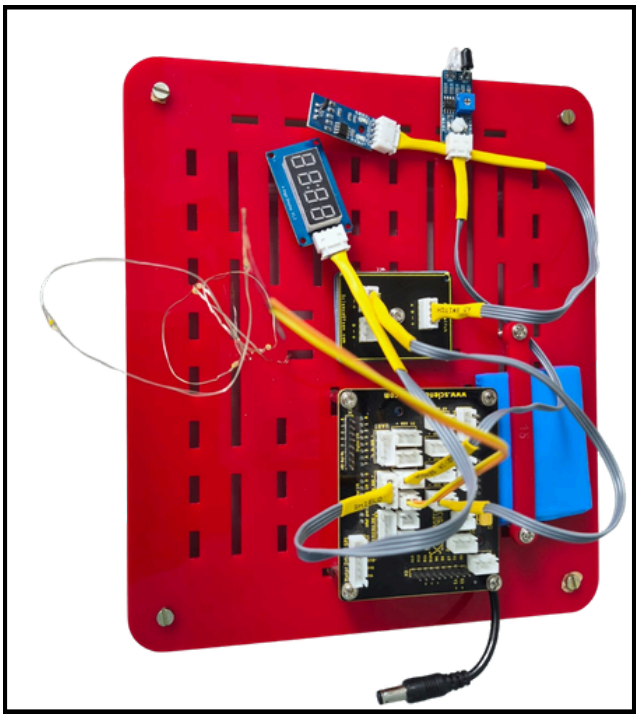
Step 3- Attach the AD switch to the base



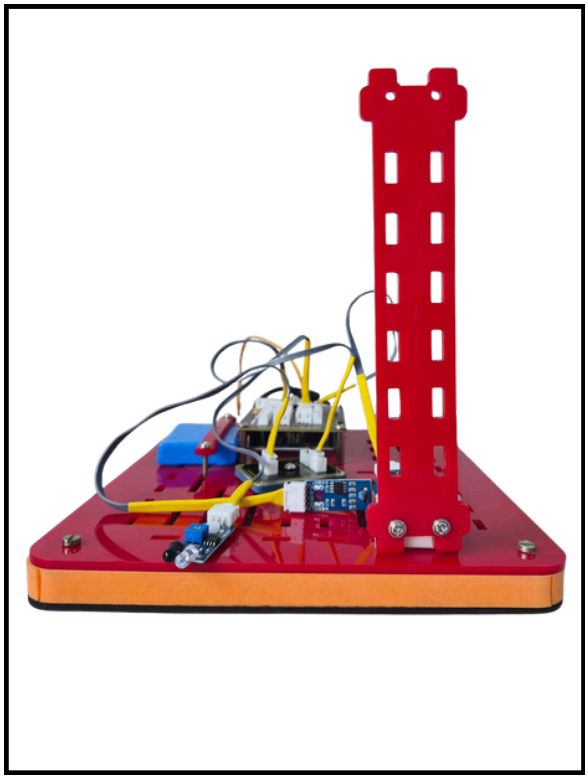
Step 4- Connect the Hall effect sensor to the AD switch



Step 5- Connect the string LED to the board

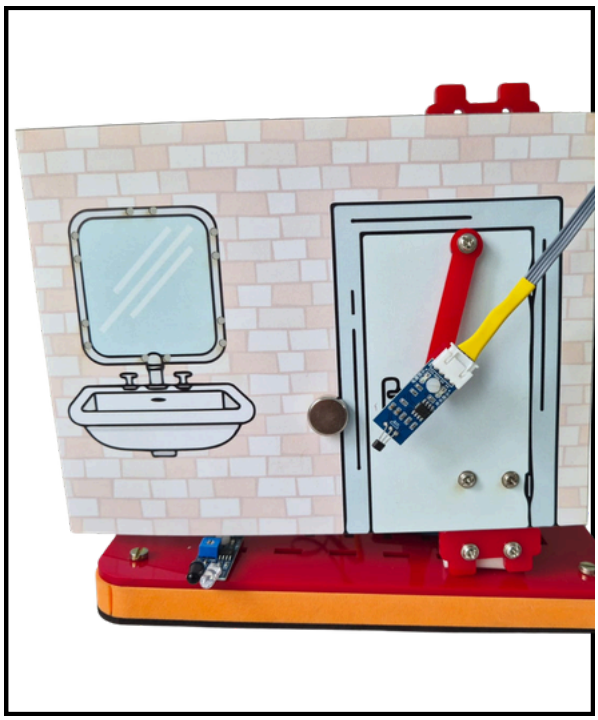


Step 6- Screw in the stand as shown

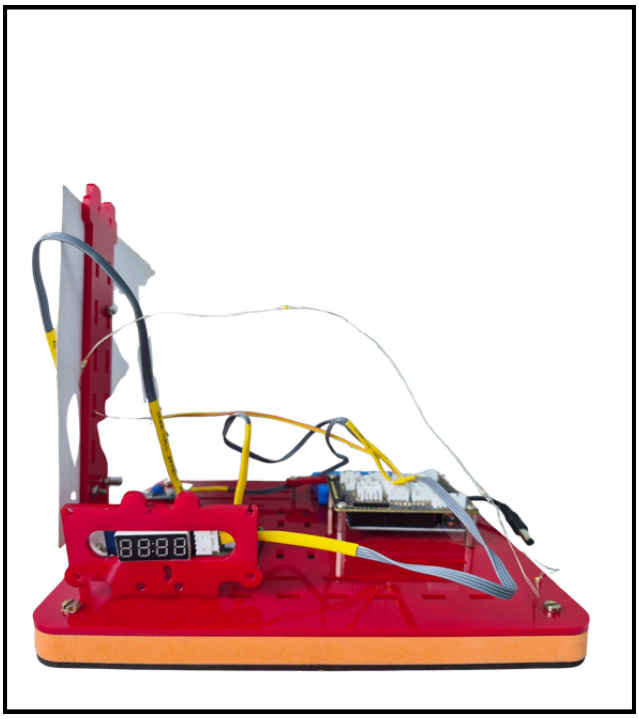


Step 7- Attach the template as shown

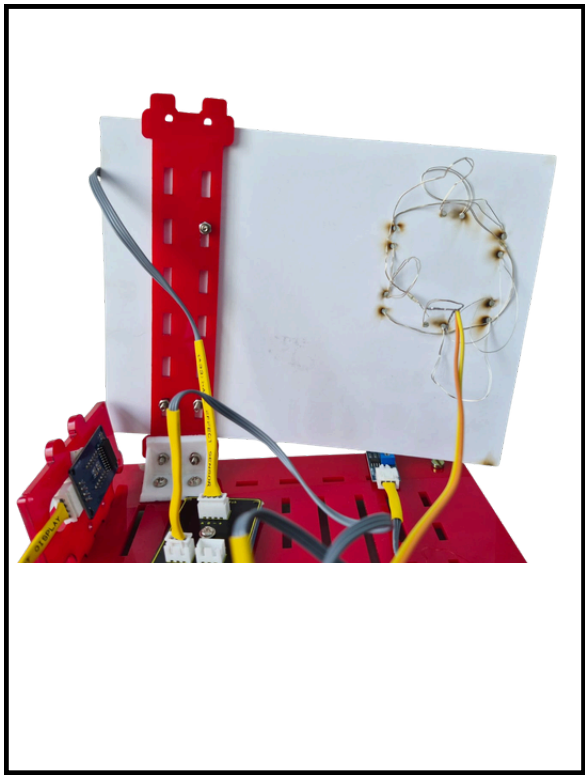
Step 8- Attach the hall effect sensor as shown



Step 9- Attach the disaplay along with it's stand to the side of the template



Step 10- Carefully sew in the string led



Step 11- Upload the code and observe

Code

```
// Session 10 Grade 9
 // Smart Bathroom Timer + Mirror Light
 // Hall sensor detects door closed and shows closed minutes on 7 segment
 // IR sensor detects person near mirror and turns string LED ON
 // Hall sensor is used as DIGITAL sensor with digitalRead(A0)

 #include <TM1637Display.h>

 const int hallSensor = A0;
 const int irSensor = 2;
 const int stringLED = 5;
 const int CLK = 9;
 const int DIO = 8;

 const int MAGNET_DETECTED = LOW; // change to HIGH if opposite
 const int IR_DETECTED = LOW;    // change to HIGH if opposite

 TM1637Display display(CLK, DIO);

 bool doorClosed = false;
 unsigned long doorCloseStart = 0;

 void setup() {
 pinMode(hallSensor, INPUT);
 pinMode(irSensor, INPUT);
 pinMode(stringLED, OUTPUT);

 display.setBrightness(7);
 display.showNumberDec(0, true);
 }
```

```
void loop() {
bool hallNow = digitalRead(hallSensor);
bool irNow = digitalRead(irSensor);

if (hallNow == MAGNET_DETECTED && !doorClosed) {
doorClosed = true;
doorCloseStart = millis();
}

if (hallNow != MAGNET_DETECTED) {
doorClosed = false;
display.showNumberDec(0, true);
}

if (doorClosed) {
unsigned long minutesClosed = (millis() - doorCloseStart) / 60000;
if (minutesClosed > 9999) minutesClosed = 9999;
display.showNumberDec(minutesClosed, true);
}

if (irNow == IR_DETECTED) {
digitalWrite(stringLED, HIGH); // mirror light ON
} else {
digitalWrite(stringLED, LOW); // mirror light OFF
}
}
```